

Agenda

Fundamental Terms and Concepts
Installing and setting up python
Literals, Variables, and Numeral Systems
Input and Output in Python (Console I/O)
Operators and Expressions

Timings:

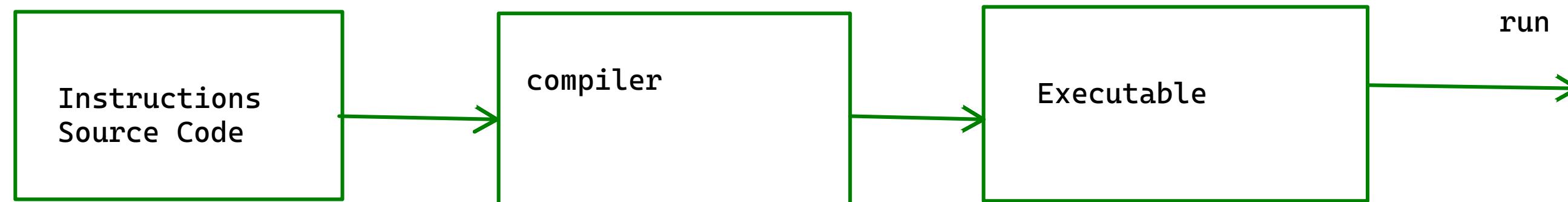
9.30AM to 11.00
15Min Break
11.15 to 1.00
1Hr Lunch break
2.00 to 3.45
15Min Break
4.00 to 5.30

Instructions in High Level language → 1/0 (Binary)

Two Ways

1. Compilation
2. Interpretation

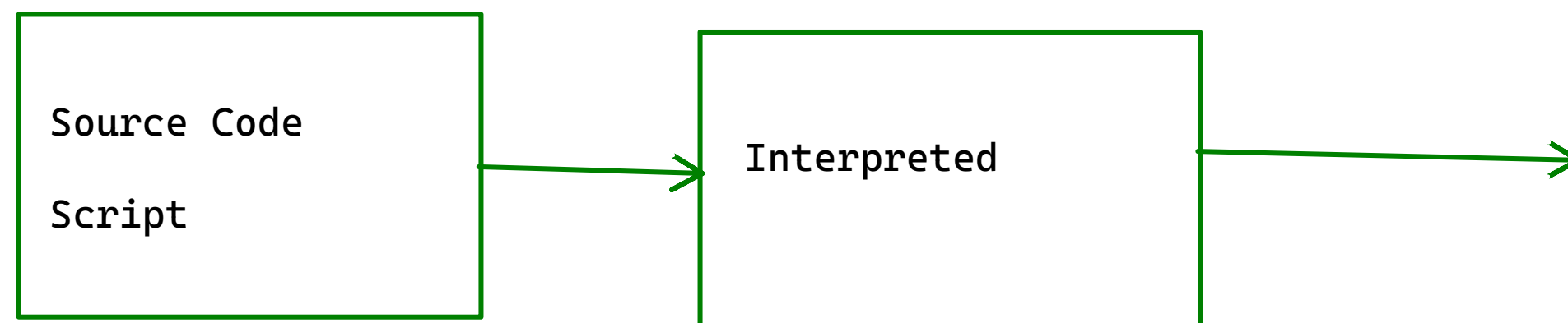
Compilation



Examples: C, C++

Linking,
Syntax (Rules for giving instructions) Check
Convert to Binary

Interpretation



Statement by Statement Execution

Examples: Python, JavaScript

Pros & Cons

Compilation:

- 1. Process takes time for compilation, Execution is faster
- 2. Any User need not have additional software to run
- 3. More granular control
- 4. much secure than interpreted languages

Interpretation:

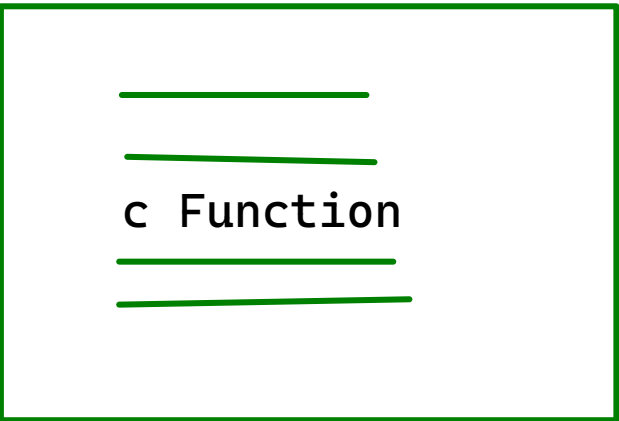
- 1. Execution of each command might take time compared to compilations
- 2. No hardware level granular control
- 3. source code (script) is shared to user
- 4. Needs an additional software (Interpreter)

Python is a
Interpreted Language
General Purpose Language
Dynamically typed language
Open Source Language
WORA – Write Once Run Any Where
Script is Independent of Arcitecture

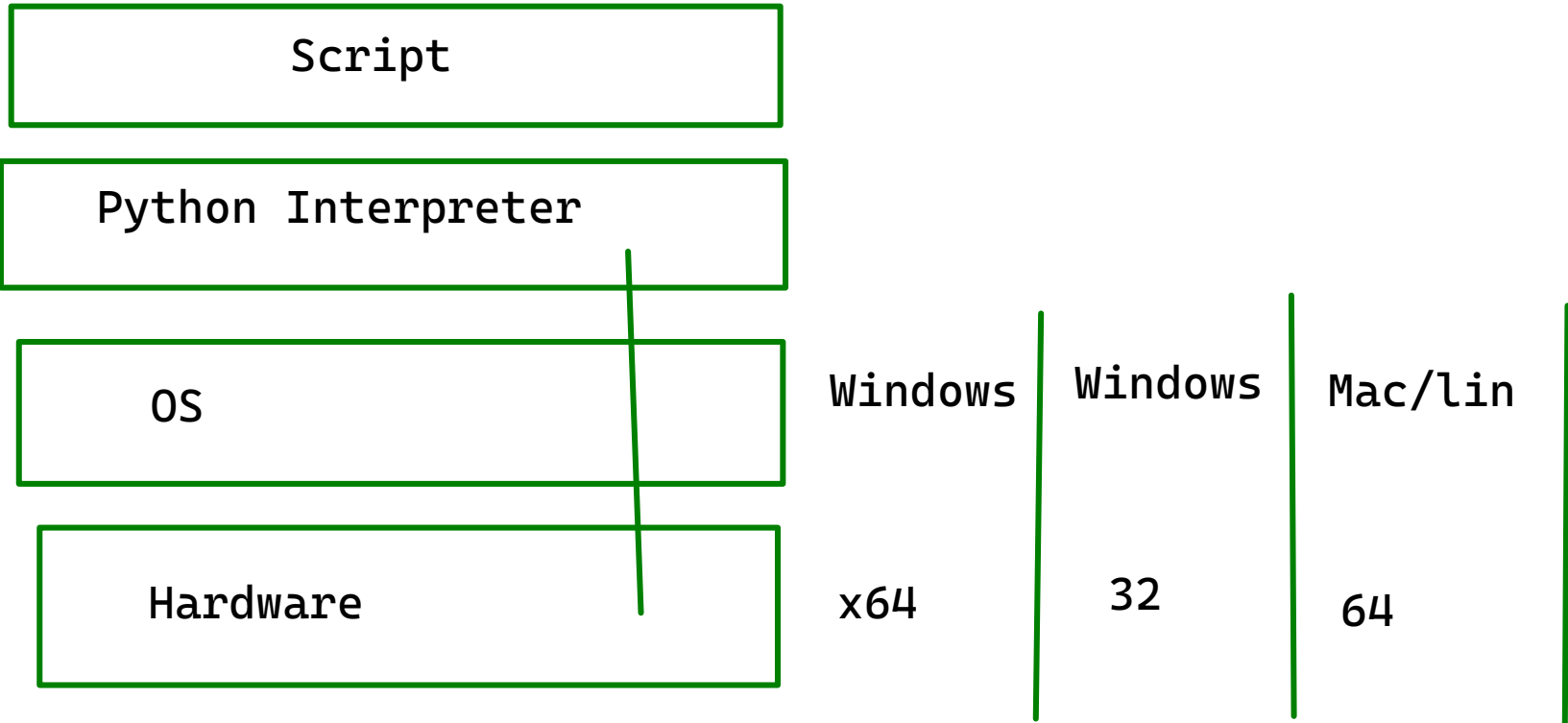
LGPL
GPL
MIT

Types of Interpreters

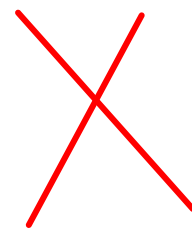
- 1. Cython
- 2. Jython
- 3. RPython
- 4. Pypi



Interpreter stack in Python



Versions of Python

2.x  3.x

Not backward Compatible

2.x

3.x (Latest)

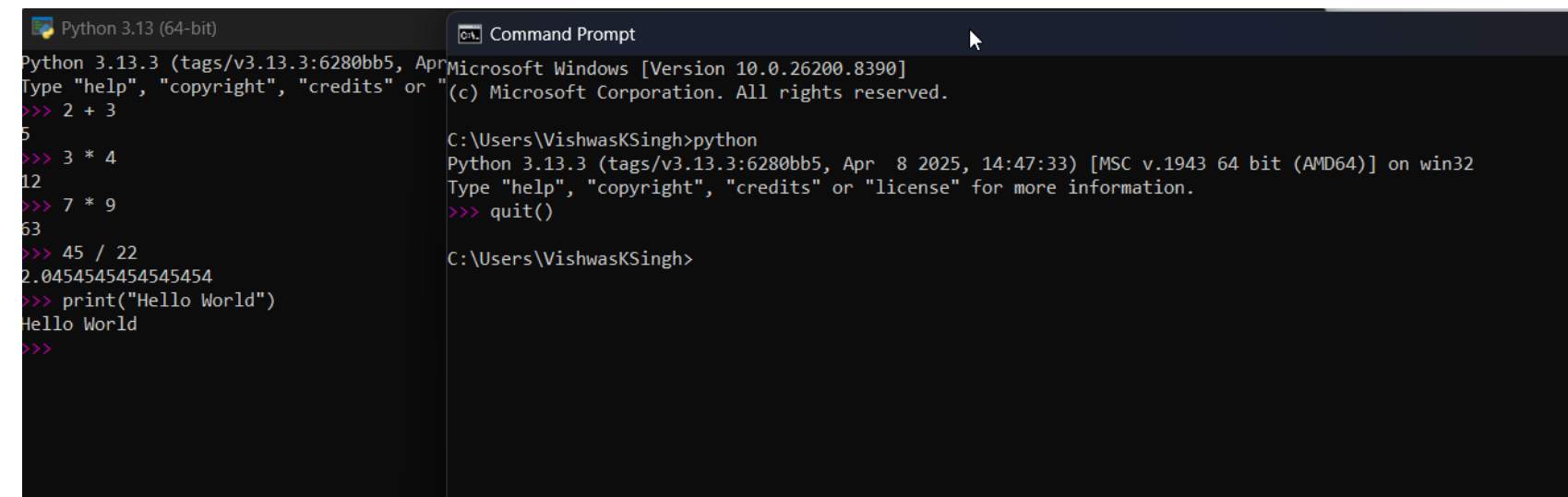
3.9 ↔ 3.10 ↔ 3.15

Things to Run & Develop a Python program

1. Python interpreter
2. Text Editor

Ways of Interacting with python interpreter

1. Interactive mode
2. Script Mode
3. Notebook mode



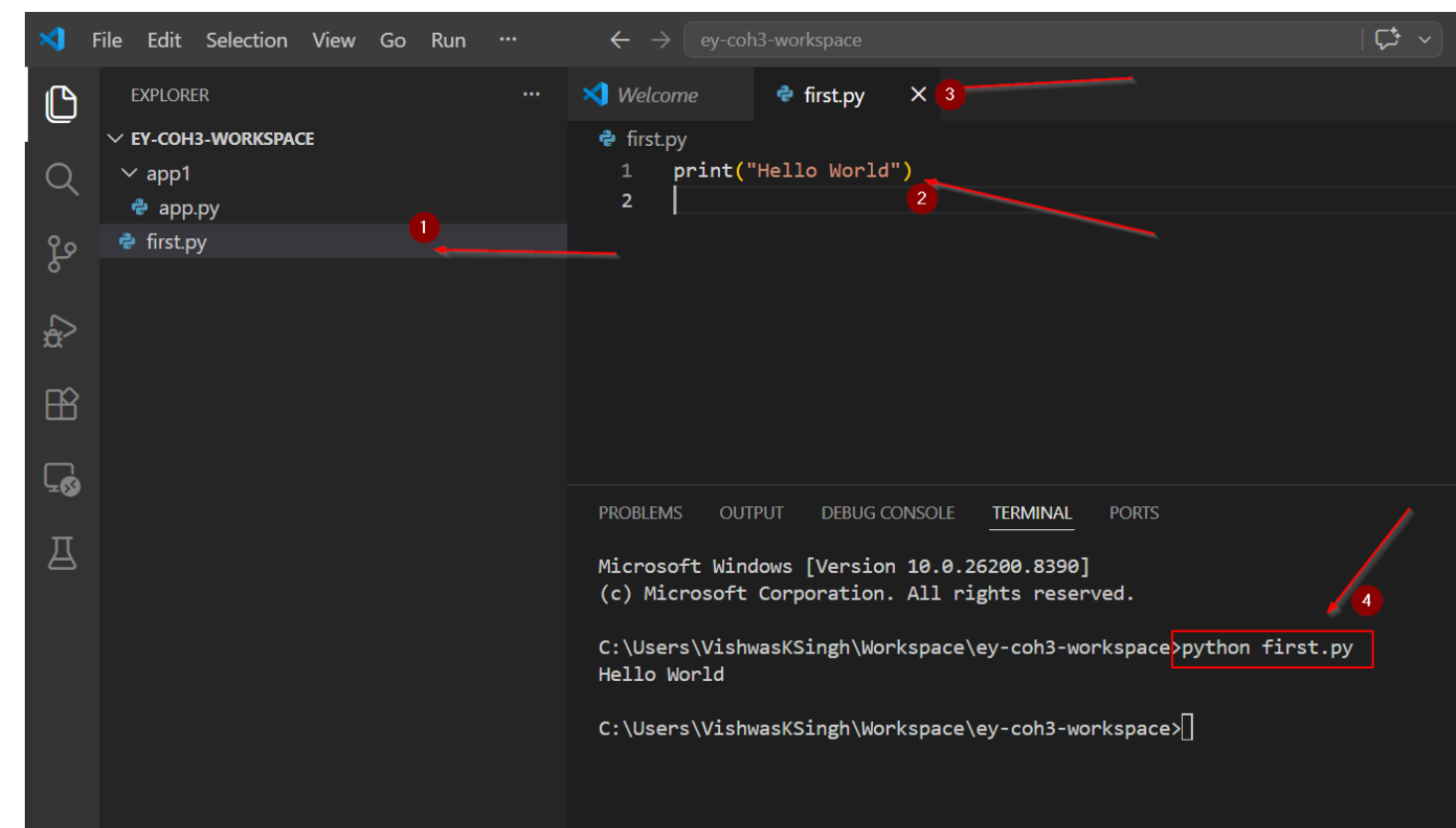
```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "quit()" for more information.
>>> 2 + 3
5
>>> 3 * 4
12
>>> 7 * 9
63
>>> 45 / 22
2.0454545454545454
>>> print("Hello World")
Hello World
>>>
```

```
Microsoft Windows [Version 10.0.26200.8390]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VishwasKSingh>python
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> quit()

C:\Users\VishwasKSingh>
```

Interactive Mode



```
File Edit Selection View Go Run ...
EXPLORER
EY-COH3-WORKSPACE
  app1
    app.py
    first.py
  first.py
  first.py

first.py
1 print("Hello World")
2

TERMINAL
Microsoft Windows [Version 10.0.26200.8390]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VishwasKSingh\Workspace\ey-coh3-workspace>python first.py
Hello World

C:\Users\VishwasKSingh\Workspace\ey-coh3-workspace>
```

Script Mode

Problem:

Analyst, Some data in webpage, Csv, Excel File, Combine those & generate a report

Sub Problems

1. Data is multiple sources - to be read

sub problem:

a. Excel File Source - Where is the File, Read the File, Send the data to next step - Function For these Steps

b. Website

2. Process it

3. Generate a report

Python Language

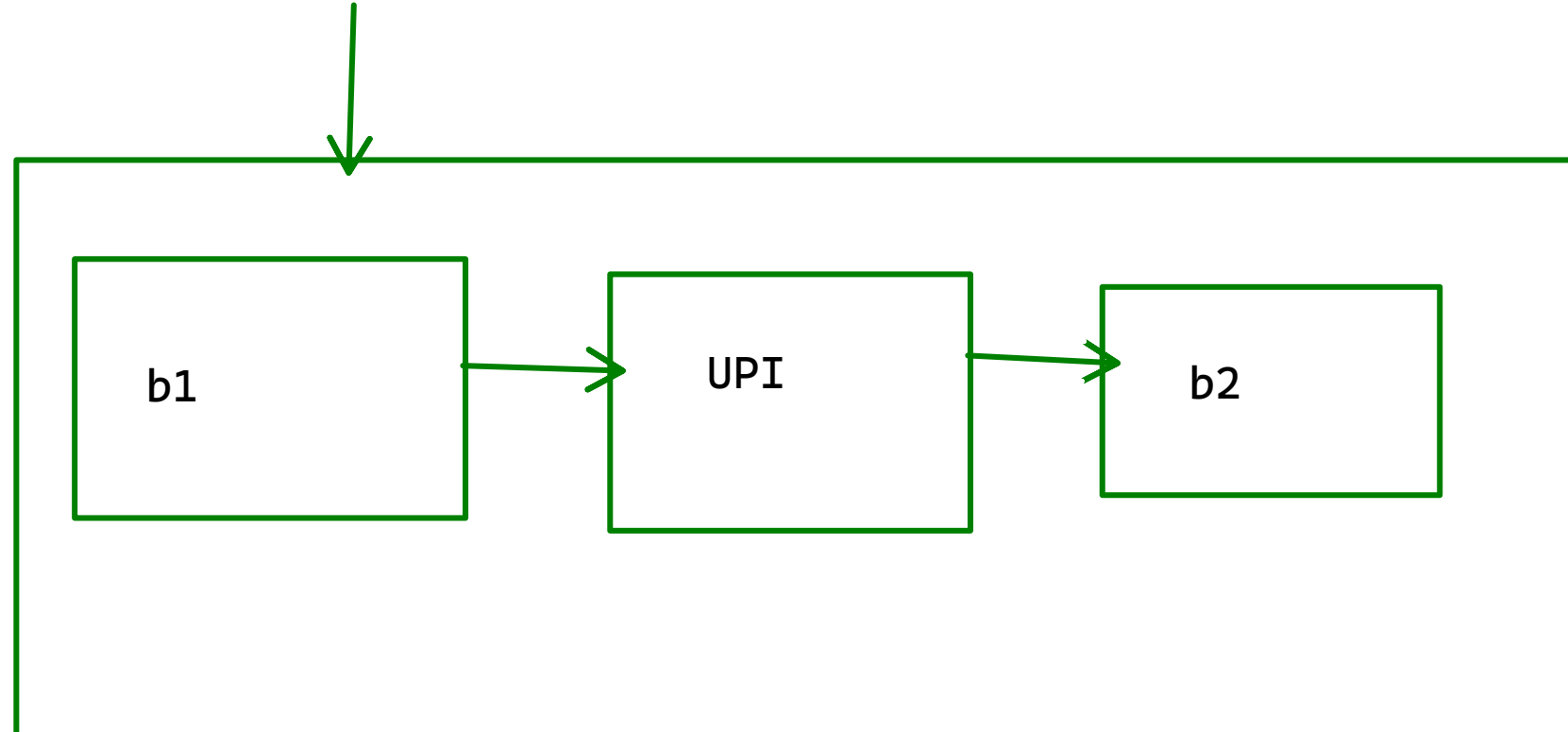
1. Why was it invented - Base Problem way
2. How to store data temporarily
3. How to do Conditions
4. How to Repetition

1. Functional Approach

but is object based

Withdraw()
Deposit()

How much amount from which bank to bank



OOPS - Object oriented Approach - Java

Banking - 100s customers

Identify the nouns - Customer, account, bank branch ...
Classes - Customer, account,

Customer	account
Cust name	acc_id
id	cust_id
acc_balance	

Can Deposit()
Can withdraw()

Create Objects -

Problem

You are saving an amount in savings bank account gives - 5% interest per annum

Input:	Output:	Logic:
principal: 300000	Interest_amount	func(p,t,r)
rate_of_interest: 5%		simple_interest = principal * rate_of_interest * time_in_months
time_in_months: 36		return simple

How to Save data temporarily:

Variables → identifier(Label) + Value

principal = 300000

Rules of a Variable:

1. It should always start with letter or _
2. It can contain number, _
3. It cannot be a key word
4. Case Sensitive

3 Types of Values: Data Type

1. Number (int, float)
2. String
3. Boolean

0x1H

principal

300000

```
>>> type_help, copyright, credits or
>>> principal = 300000
>>> type(principal)
<class 'int'>
>>> rate_of_interest = 0.07
>>> type(rate_of_interest)
<class 'float'>
>>> nme = "vishwas"
>>> type(nme)
<class 'str'>
>>> isValid = True
>>> type(isValid)
<class 'bool'>
>>>
```

```
>>> cp = complex(2,3)
>>> cp
(2+3j)
>>> type(cp)
<class 'complex'>
>>> hexnum = 0x1A
>>> type(hexnum)
<class 'int'>
>>> hexnum
26
>>> binnum = 0b1010
>>> binnum
10
>>> type(binnum)
<class 'int'>
>>> k = 152364575937264845756383628274628263747562826473826384632628374638263846382638463836482628473827393748478273645\
85758459374
>>> k
152364575937264845756383628274628263747562826473826384632628374638263846382638463836482628473827393748478273645857584593
74
>>> k1 = 6.3e10
>>> k1
63000000000.0
>>> type(k1)
<class 'float'>
>>> p = 20_00_000
>>> p
2000000
>>> |
```

Conventions of Naming:

1. Always have meaningful names
2. variables with multiple words follow snake_case

Operators in Python

Any number which is other than 0 is always true

Zero is always false

The input taken from a user is always a string, developer has to convert to right type

Print Function

1. Can take any number of values
2. By default it inserts a new line
3. It inserts a space as a separator by default , it can be overridden by sep="Value"

format specifiers: %d, %f, %s etc ...

Once a character # appears in a statement – the sequence following it is considered a comment

Comments are not processed

Comments are for developers or users who use your code

Standard to maintain readability PEP8 Style Guide

Linting Process – pylint(pep8), pylance, flake8